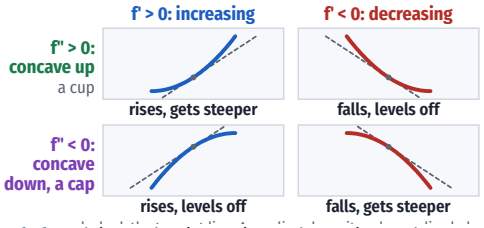
1 The four shapes from the signs of f' and f''

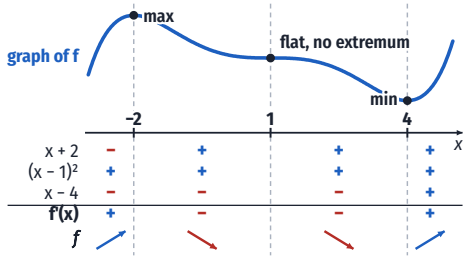


Label: each interval as increasing or decreasing, and as concave up or concave down.

1. The sign of f' says rising (plus) or falling (minus). The sign of f'' says cup (plus) or cap (minus).
2. Pick the matching piece on each interval and join the pieces at the special points.

- A relative max is the top of a cap, a relative min is the bottom of a cup, and an inflection point is where the bend flips.
- On a graph of f': f rises where f' is above the x-axis, and f is concave up where f' is rising.

2 Sign chart on a factored f'



Label: the break numbers, one row per factor, the row for f', and an arrow for f in each interval.

1. The break numbers are the zeros of each factor and the numbers not in the domain.
2. Test one value inside each interval, write the sign of each factor, and multiply down the column.

- Check one column with numbers: f'(0) = (2)(1)(−4) = −8, which matches the minus sign.
- $(x - 1)^2$ is never negative, so f' keeps its sign at 1 and there is no extremum there. Factors like e^x and $x^2 + 1$ are always positive.

3 First derivative test: max, min, or neither

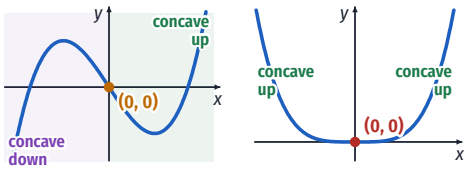


Label: the critical number c, the sign of f' on each side of it, and the point (c, f(c)).

1. A critical number is a c in the domain where f'(c) = 0 or f'(c) does not exist.
2. Read the sign of f' on each side of c, and compute f(c) from f itself.

- Plus then minus is a relative max. Minus then plus is a relative min. No sign change means no extremum.
- $f(x) = x^{2/3}$ has $f'(x) = 2 \div (3x^{1/3})$, which does not exist at 0, so 0 is a critical number.

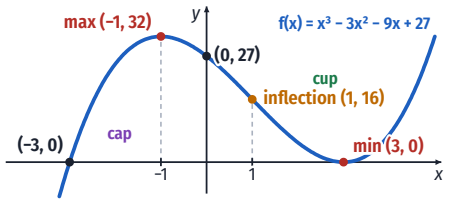
4 Concavity and inflection points



Label: where the graph is concave up and concave down, and each inflection point as (c, f(c)).

1. Factor f'' and make its sign chart. Plus means concave up and minus means concave down.
2. An inflection point needs f'' to change sign at a c where f is continuous.

5 Curve sketching checklist, one full example

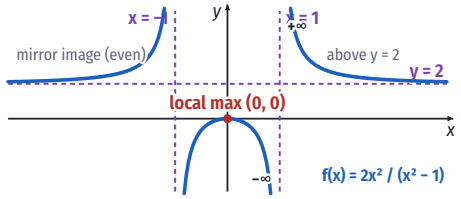


Label: the intercepts, every extremum and inflection point with its coordinates, and any asymptotes dashed.

1. Domain, intercepts, symmetry, and asymptotes. Here f = (x − 3)²(x + 3), so the graph crosses at −3 and touches at 3.
2. f' = 3(x − 3)(x + 1) gives the max (−1, 32) and the min (3, 0).
3. f'' = 6(x − 1) gives the inflection point (1, 16).
4. Fill in the table, plot the points, and join them with the shapes from card 1.

Interval	(−∞, −1)	(−1, 1)	(1, 3)	(3, ∞)
f' and f''	+ and −	− and −	− and +	+ and +
Shape	rising cap	falling cap	falling cup	rising cup

6 Asymptotes of a rational function

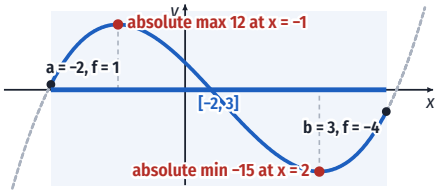


Label: the dashed lines x = −1, x = 1, and y = 2 with their equations, and the local max (0, 0).

1. Cancel common factors. Each zero of the denominator that is left gives a vertical asymptote. A cancelled factor gives a hole.
2. Near x = 1 the top is about 2 and the bottom is tiny: positive on the right side and negative on the left.
3. Divide the top and the bottom by x^2. The limit at both ends is 2.

- A smaller top degree gives y = 0. Equal degrees give the ratio of the leading coefficients. A bigger top degree gives no horizontal asymptote.

7 Absolute extrema on a closed interval

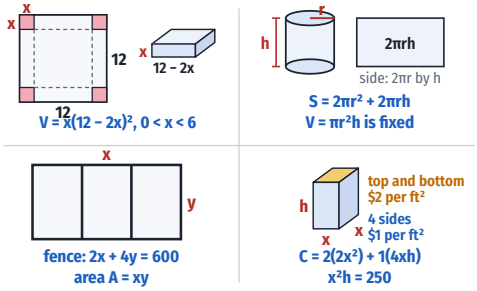


Label: both endpoints, the critical numbers inside, and the largest and smallest values.

1. f(x) = 2x³ − 3x² − 12x + 5 is continuous on [−2, 3], so both absolute extrema exist.
2. f'(x) = 6(x − 2)(x + 1) gives the critical numbers −1 and 2. Both are inside.
3. Make the value table and pick the largest and the smallest value.

x	−2 (end)	−1	2	3 (end)
f(x)	1	**12, max**	**−15, min**	−4

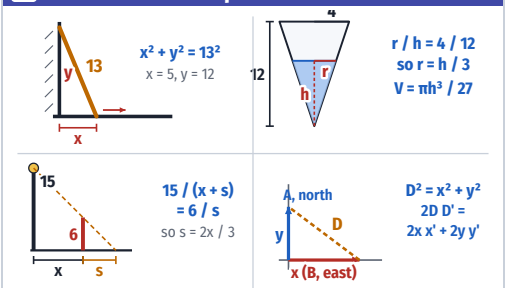
8 Optimization: four setups



Label: a letter on every length with units, the objective, the constraint, and the domain.

- Open box: V' = (12 − 2x)(12 − 6x) = 0 gives x = 2 in (0, 6), so V = 128 cubic inches.
- Can: the least material is at h = 2r. Pen: x = 150 ft and y = 75 ft. Box: x = 5 ft, h = 10 ft, and the cost is $300.

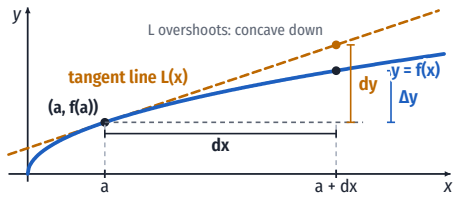
9 Related rates: four pictures



Label: changing lengths as letters, fixed lengths as numbers, and each rate with its sign.

- Cars at 60 and 80 mph: after 3 minutes y = 3 and x = 4, so D = 5 and D' = (4 · 80 + 3 · 60) ÷ 5 = 100 mph.
- Shadow: s = 2x/3, so the shadow grows at 2/3 of the walking speed.

10 Linear approximation: tangent line versus curve

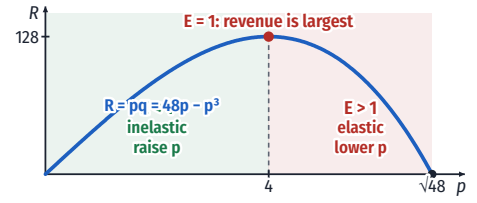


Label: the base point a, the run dx, dy up to the tangent line, and Δy up to the curve.

1. Pick a near x where f(a) and f'(a) are exact. Then f(x) ≈ f(a) + f'(a) dx, with dx = x − a.

- Concave down puts the tangent line above the curve, so the estimate is too big: √26 ≈ 5 + (1/10)(1) = 5.1 is a little high.

11 Elasticity and revenue

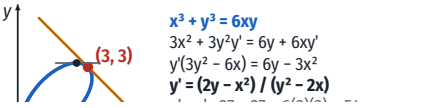


Label: the price where E = 1 at the top of R, the inelastic side, and the elastic side.

1. q = 48 − p² gives E = 2p² ÷ (48 − p²). E = 1 when 2p² = 48 − p², so at p = 4.

- dR/dp = q(1 − E), so revenue rises with the price when E < 1 and falls when E > 1.

12 Implicit curve and its tangent line



Math 16A Midterm 2 · Side 2 · Every rule and formula

Review sheet: memorize this, it is not allowed in the exam.

Derivative rules

Rule	Formula	Example
Power, any n	$(x^n)' = nx^{n-1}$	$(\sqrt{x})' = \frac{1}{2\sqrt{x}}$
Product	$(uv)' = u'v + uv'$	$(x^2e^x)' = 2xe^x + x^2e^x$
Quotient	$\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}$	$\left(\frac{x}{x+1}\right)' = \frac{1}{(x+1)^2}$
Chain	$[f(g(x))]' = f'(g(x))\,g'(x)$	$[(3x+1)^5]' = 15(3x+1)^4$
Power and chain	$(g^n)' = ng^{n-1}g'$	$(\sqrt{x^2+9})' = \frac{x}{\sqrt{x^2+9}}$

f	f'	With inside u	f	f'	With inside u
e^x	e^x	$e^u\,u'$	$\sin x$	$\cos x$	$\cos u \cdot u'$
a^x	$(\ln a)a^x$	$(\ln a)a^u\,u'$	$\cos x$	$-\sin x$	$-\sin u \cdot u'$
$\ln x$	$\frac{1}{x}$	$\frac{u'}{u}$	$\tan x$	$\sec^2 x$	$\sec^2 u \cdot u'$
$\log_a x$	$\frac{1}{(\ln a)x}$	$\frac{u'}{(\ln a)u}$	$\cot x$	$-\csc^2 x$	$-\csc^2 u \cdot u'$
$\ln\lvert x\rvert$	$\frac{1}{x}$	$\frac{u'}{u}$	$\sec x$	$\sec x \tan x$	$\sec u \tan u \cdot u'$
k	0	$(kf)' = kf'$	$\csc x$	$-\csc x \cot x$	$-\csc u \cot u \cdot u'$

The three functions whose names start with "co" (cosine, cotangent, cosecant) are the three with a minus sign.

- The power rule needs a constant exponent, as in x^5. The a^x rule needs a constant base, as in 5^x. For x^x, take ln of both sides first: the derivative is $x^x(\ln x + 1)$.
- $\sin^2 x$ means $(\sin x)^2$, so its derivative is $2 \sin x \cos x$. The derivative of $\sin(x^2)$ is $2x\cos(x^2)$.
- With three layers, write one factor per layer from the outside in: the derivative of $\ln(\cos(e^{2x}))$ is $[-\sin(e^{2x}) \cdot 2e^{2x}] \div \cos(e^{2x})$.
- Numbers such as e^2, ln 3, and $\cos(\pi/6)$ are constants, so each has derivative 0.

Exponents and logarithms

$\ln(xy) = \ln x + \ln y$, $\ln(x/y) = \ln x - \ln y$, $\ln(x^r) = r \ln x$, and $\log_a x = \ln x \div \ln a$.

- $\ln 1 = 0$, $\ln e = 1$, $e^{\ln x} = x$, $\ln(e^x) = x$, and $a^x = e^{x \ln a}$.
- $x^a x^b = x^{a+b}$, $x^a / x^b = x^{a-b}$, $(x^a)^b = x^{ab}$, $x^{-a} = 1/x^a$, and $x^{m/n}$ is the n-th root of x^m.
- ln x exists only for $x > 0$. A power of e is always positive. To solve $e^x = 5$, take ln of both sides: $x = \ln 5$.
- Simplify a log before you differentiate: $\ln(x^3 \div \sqrt{(x+1)}) = 3 \ln x - \frac{1}{2}\ln(x+1)$, whose derivative is $3/x - 1/(2(x+1))$.
- TRAP $\ln(x^2 + 5)$ cannot be split. Logs split only products, quotients, and powers.

Trig values and identities

Angle	0	$\frac{\pi}{6}$ (30°)	$\frac{\pi}{4}$ (45°)	$\frac{\pi}{3}$ (60°)	$\frac{\pi}{2}$ (90°)
sin	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
cos	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
tan	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	undefined

- $\tan x = \sin x / \cos x$, $\cot x = \cos x / \sin x$, $\sec x = 1/\cos x$, and $\csc x = 1/\sin x$.
- $\sin^2 x + \cos^2 x = 1$, $1 + \tan^2 x = \sec^2 x$, and $\sin 2x = 2 \sin x \cos x$. Use radians: π radians is 180°.

For other angles, the first-quadrant reference angle gives the size and the quadrant gives the sign. All are positive in quadrant I, only sin in II, only tan in III, and only cos in IV.

- Example: $5\pi/6$ has reference angle $\pi/6$ in quadrant II, so $\sin(5\pi/6) = 1/2$ and $\cos(5\pi/6) = -\sqrt{3}/2$. On $[0, 2\pi]$, $\sin x = 1/2$ at both $\pi/6$ and $5\pi/6$.

L'Hospital's rule

If substituting gives 0/0 or ±∞/±∞, then the limit of f/g equals the limit of f'/g' (when that limit exists or is ±∞).

1. Substitute, and write the limit of the top and of the bottom. Name the form in words.
2. If the form is not 0/0 or ∞/∞, stop: the substitution answer is the limit.
3. Differentiate the top and the bottom separately (no quotient rule), then substitute again. Repeat if needed.

- Rewrite $0 \cdot \infty$ as a quotient first, as in $x \ln x = (\ln x) \div (1/x)$. Combine $\infty - \infty$ into one fraction first.
- To know: $\lim_{x\to 0} \frac{\sin x}{x} = 1$, $\lim_{x\to 0} \frac{e^{3x}-1}{x} = 3$, $\lim_{x\to\infty} \frac{\ln x}{x} = 0$, $\lim_{x\to\infty} \frac{x^2}{e^x} = 0$, $\lim_{x\to 0^+} x \ln x = 0$.
- For large x, ln x grows more slowly than any power x^p with $p > 0$, and every power grows more slowly than e^x.
- TRAP $(x + \cos x)/(x + 2)$ as x goes to 0 has the form 1/2, which is not indeterminate. The limit is 1/2, so do not use the rule.

Definitions to be able to state

A critical number of f is a number c in the domain of f where f'(c) = 0 or f'(c) does not exist.

- An inflection point is a point (c, f(c)) on the graph, with f continuous at c, where the concavity changes.
- Extreme value theorem: a function continuous on a closed interval [a, b] has an absolute max and an absolute min on [a, b].
- The tangent line at x = a is $y - f(a) = f'(a)(x - a)$.
- Velocity v = s' and acceleration a = v' = s". An object speeds up when v and a have the same sign.
- Marginal cost is C'(x), and the same for R and P. Revenue R = xp, profit P = R − C, and average cost is C(x)/x.
- The point of diminishing returns is the inflection point where the graph goes from concave up to concave down.

Sign chart and first derivative test

1. Find the domain. Write f' as one fraction and factor its top and bottom completely.
2. Critical numbers are numbers in the domain where the top or the bottom of f' is 0.
3. Put them and the numbers outside the domain on a line, and test one value inside each interval.
4. f' > 0 means increasing and f' < 0 means decreasing. Answer with open intervals, never joined across a number outside the domain.
5. Plus to minus is a relative max, minus to plus is a relative min, and no change is neither. Get f(c) from f.

Second derivative test

- At a c with f'(c) = 0: if f"(c) > 0, the graph is a cup, so f has a relative min. If f"(c) < 0, it is a cap, so f has a relative max.
- If f"(c) = 0, or f'(c) does not exist, this test says nothing. Use the first derivative test.

Absolute extrema

Closed interval [a, b]: the closed interval method

1. Say f is continuous on [a, b], so the extreme value theorem says the absolute max and min exist.
2. Find the critical numbers, and throw out in writing any that are not in (a, b).
3. Evaluate f at the critical numbers left and at a and b. The largest value is the absolute max and the smallest is the absolute min. Give each value and the x where it happens.

Other intervals: the critical point theorem

4. If f is continuous on the interval and c is its only critical number there, a relative min at c is the absolute min, and a relative max at c is the absolute max.
5. For the other kind, look near the ends: if f(x) goes to ∞, there is no absolute max.

- Example: $f(x) = x + 9/x$ on $(0, \infty)$ has $f'(x) = (x^2 - 9)/x^2$, so 3 is the only critical number, and $f"(3) = 18/27 > 0$. The absolute min is f(3) = 6. There is no absolute max, because f(x) goes to ∞ as x approaches 0 from the right.

Optimization

1. Draw and label a picture. Define each variable in a sentence with units.
2. Write the objective as a formula and the constraint as an equation. Solve the constraint for one variable and substitute.
3. State the domain from the context, such as $x > 0$ or $0 \le x \le 6$.
4. Differentiate and find the critical numbers in the domain.
5. Justify the max or min: the closed interval method on a closed, bounded domain, or the critical point theorem with a derivative test on an open one.
6. Answer in a sentence with units, giving every dimension asked for.

- Example: square base, volume 250, top and bottom at $2 per square foot, sides at $1. With $h = 250/x^2$, $C(x) = 4x^2 + 1000/x$ and $C'(x) = 8x - 1000/x^2 = 0$ gives $x^3 = 125$, so x = 5 and h = 10. Since $C"(x) = 8 + 2000/x^3 > 0$ and 5 is the only critical number, the least cost is C(5) = $300.

Elasticity of demand and revenue

E = −(p/q)(dq/dp). In this course E is always positive.

1. Write q as a function of p, and find dq/dp as a formula before putting in any price.
2. Substitute into E and simplify, then put in the price and compare E with 1.

E	Demand is	To raise revenue
E < 1	inelastic	raise the price (price and revenue move together)
E = 1	unit elastic	nothing: revenue is at its max

Economic order quantity (lot size)

- M units a year come in batches of q. Each batch costs f to set up, each unit costs g, and storing one unit for a year costs k.

C(q) = fM/q + gM + kq/2 for $0 < q \le M$. Then $C'(q) = -fM/q^2 + k/2 = 0$ gives $q = \sqrt{(2fM/k)}$.

- The kq/2 is there because the average inventory is half a batch. There are M/q batches a year, and g does not change the best q.
- Justify: $C"(q) = 2fM/q^3 > 0$ for $q > 0$, and q is the only critical number, so it gives the absolute minimum cost.
- TRAP If storage is charged on a full batch (rented space), rebuild C(q) from its parts with kq instead of kq/2.

Implicit differentiation

1. Differentiate both sides with respect to x, treating y as a function of x: $(y^3)' = 3y^2y'$, $(e^y)' = e^y y'$, $(xy)' = y + xy'$.
2. Move the y' terms to one side and the rest to the other. Factor out y' and divide.
3. For a slope at a point, check the point is on the curve, then substitute both coordinates.

- Example: $x^2 + y^2 = 25$ gives $2x + 2yy' = 0$, so $y' = -x/y$. At (3, 4), which is on the circle since 9 + 16 = 25, the slope is −3/4.

Related rates

1. Draw a picture. Changing lengths get letters, and fixed lengths get numbers.
2. Name each variable with units at time t. Write the given and wanted rates as d/dt. A shrinking quantity has a negative rate.
3. Write an equation that is true at every time, and differentiate both sides with respect to t.
4. Only now substitute the numbers for the instant. Find missing values from the original equation.
5. Solve, and answer with units, saying whether the quantity is increasing or decreasing.

- Ladder, 13 ft, bottom pulled out at 2 ft/s, bottom 5 ft out: $2x(dx/dt) + 2y(dy/dt) = 0$ gives $2(5)(2) + 2(12)(dy/dt) = 0$, so $dy/dt = -5/6$. The top slides down at 5/6 ft/s.
- Cone tank 12 ft tall, top radius 4, draining 2 cubic ft per minute: $V = \pi h^3/27$, so $dV/dt = (\pi h^2/9)(dh/dt)$. At h = 6, $-2 = 4\pi(dh/dt)$, so the level falls at $1/(2\pi)$ ft per minute.
- Sphere: $V = (4/3)\pi r^3$ gives $dV/dt = 4\pi r^2(dr/dt)$.

Linear approximation and error

$L(x) = f(a) + f'(a)(x - a)$. The differential $dy = f'(x)\,dx$ approximates the true change $\Delta y = f(x + dx) - f(x)$.

- Error in Q from a measured x with maximum error dx: $dQ = Q'(x)\,dx$, at the measured x, in the units of Q.

Relative error = dQ/Q. Percentage error = 100 × dQ/Q. Neither has units.

- If you are given a diameter but the formula uses the radius, halve both the value and the error.
- Example: r = 6 cm with error 0.02 cm gives $dV = 4\pi r^2 dr = 4\pi(36)(0.02) = 2.88\pi$ cubic cm. Since $V = 288\pi$, the relative error is 0.01, or 1%.

Geometry formulas

Shape	Area or volume	Perimeter or surface area
Rectangle, triangle	lw, ½bh	2l + 2w; add the sides
Circle	πr^2	circumference $2\pi r$
Box with square base	x^2h	closed $2x^2 + 4xh$; open top $x^2 + 4xh$
Cylinder	$\pi r^2 h$	closed $2\pi r^2 + 2\pi rh$; side $2\pi rh$
Cone	$(1/3)\pi r^2 h$	side $\pi r\sqrt{(r^2 + h^2)}$
Sphere	$(4/3)\pi r^3$	$4\pi r^2$

- Right triangle: $a^2 + b^2 = c^2$. To minimize a distance $\sqrt{((x-a)^2 + (y-b)^2)}$, minimize its square.

Sentences that earn the justification points

- "f has a relative max at x = c because f' changes from positive to negative at c."
- "f'(0) = 0, but f" does not change sign at 0, so there is no inflection point at x = 0."
- "f is continuous on the closed interval [a, b], so by the extreme value theorem it has an absolute max and min."
- "The critical number x = 5 is not in [0, 4], so I discard it."
- "x = c is the only critical number in (0, ∞) and C"(c) > 0, so by the critical point theorem it gives the absolute minimum."
- "The top and the bottom both approach 0, so the form is 0/0 and L'Hospital's rule applies."

Most common mistakes

- TRAP Forgetting the inside derivative. The derivative of e^{-2x}